Experiment-13

Write a python program to implement Minimax algorithm for gaming

Aim

To write and execute a Python program to implement the Minimax algorithm to find the optimal value for the MAX player in a game tree.

Algorithm

1. Represent the game tree of depth 3 using the leaf node values.
2. Treat the root node as the MAX player and alternate between MAX and MIN at each level.
3. If the node is a leaf node, return its value.
4. Recursively evaluate the left child and the right child of the node.
5. At a MAX node, choose the maximum of the child values.
6. At a MIN node, choose the minimum of the child values.
7. Pass the selected value up to the parent node until the root is reached.
8. Display the optimal value at the root node.

Coding

ex-13.py - D:/ex-13.py (3.10.0)

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```
# Minimax Algorithm for gaming
import math

# Leaf node values of a game tree of depth 3
scores = [3, 5, 6, 9, 1, 2, 0, -1]

def minimax(depth, index, is_max, height):

    # Leaf node : return its score
    if depth == height:
        return scores[index]

    left = minimax(depth + 1, index * 2, not is_max, height)
    right = minimax(depth + 1, index * 2 + 1, not is_max, height)

    if is_max:
        value = max(left, right)
        print("MAX node at depth", depth, ":", left, right, "->", value)
    else:
        value = min(left, right)
        print("MIN node at depth", depth, ":", left, right, "->", value)
    return value

height = int(math.log2(len(scores)))

print("MINIMAX ALGORITHM")
print("-----------------")
print("Leaf values :", scores)
print()
best = minimax(0, 0, True, height)
print()
print("The optimal value is :", best)
```

Output

IDLE Shell 3.10.0

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```
Python 3.10.0 (tags/v3.10.0:b494f59, Oct  4 2021, 19:00:18) [MSC v.1929 64 bit (
AMD64)] on win32
Type "help", "copyright", "credits" or "license()" for more information.
>>>
========================== RESTART: D:/ex-13.py ==========================
MINIMAX ALGORITHM
-----------------
Leaf values : [3, 5, 6, 9, 1, 2, 0, -1]

MAX node at depth 2 : 3 5 -> 5
MAX node at depth 2 : 6 9 -> 9
MIN node at depth 1 : 5 9 -> 5
MAX node at depth 2 : 1 2 -> 2
MAX node at depth 2 : 0 -1 -> 0
MIN node at depth 1 : 2 0 -> 0
MAX node at depth 0 : 5 0 -> 5

The optimal value is : 5
>>>
```

Result

The Minimax algorithm was successfully implemented. For the given leaf values, the optimal value for the MAX player at the root node is 5.